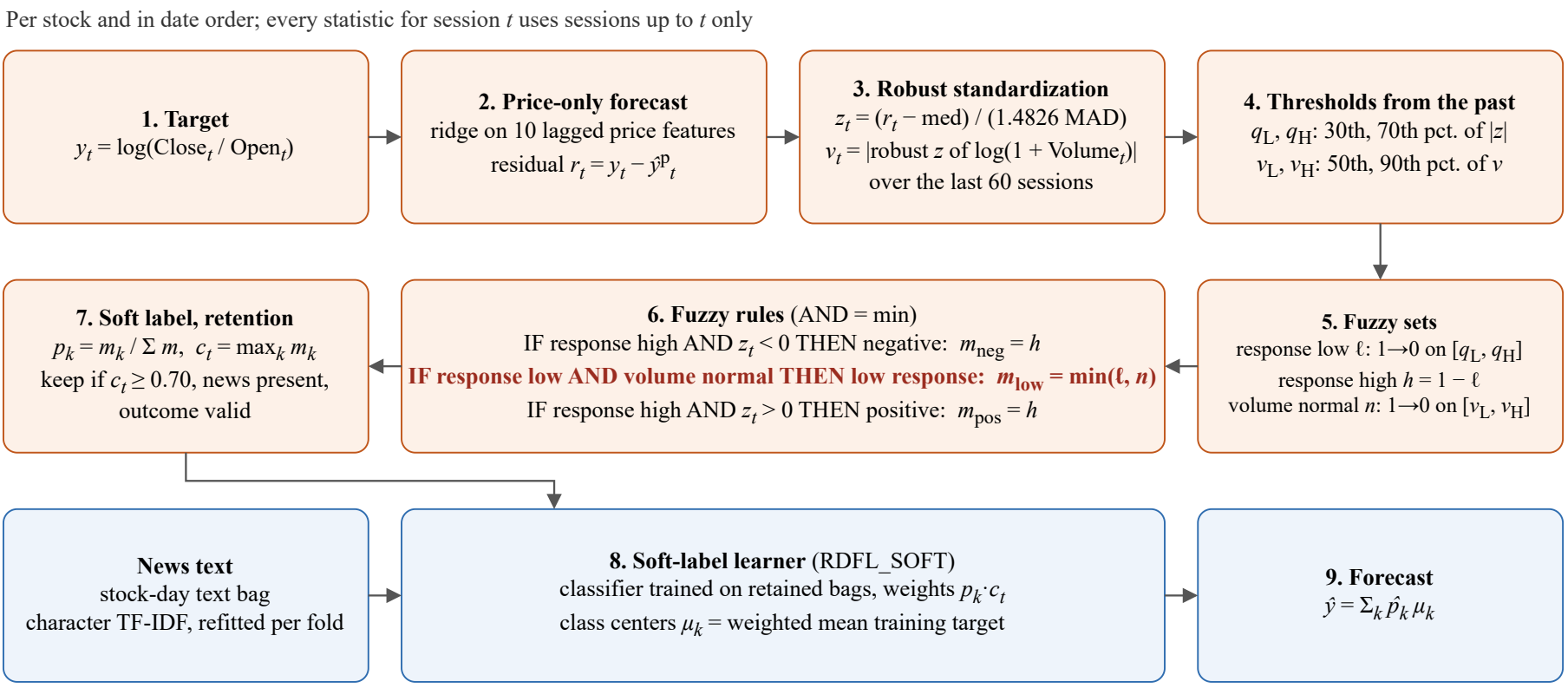

Per stock and in date order; every statistic for session $t$ uses sessions up to $t$ only
1. Target
$y_t = \log(\text{Close}_t / \text{Open}_t)$
2. Price-only forecast
ridge on 10 lagged price features
residual $r_t = y_t - \hat{y}^{p}_t$
3. Robust standardization
$z_t = (r_t - \text{med}) / (1.4826\ \text{MAD})$
$v_t = |\text{robust } z \text{ of } \log(1 + \text{Volume}_t)|$
over the last 60 sessions
4. Thresholds from the past
$q_L$, $q_H$: 30th, 70th pct. of $|z|$
$v_L$, $v_H$: 50th, 90th pct. of $v$
5. Fuzzy sets
response low $\ell$: 1→0 on $[q_L, q_H]$
response high $h = 1 - \ell$
volume normal $n$: 1→0 on $[v_L, v_H]$
6. Fuzzy rules (AND = min)
IF response high AND $z_t < 0$ THEN negative: $m_{neg} = h$
IF response low AND volume normal THEN low response: $m_{low} = \min(\ell, n)$
IF response high AND $z_t > 0$ THEN positive: $m_{pos} = h$
7. Soft label, retention
$p_k = m_k / \Sigma\, m$, $c_t = \max_k m_k$
keep if $c_t \geq 0.70$, news present, outcome valid
News text
stock-day text bag
character TF-IDF, refitted per fold
8. Soft-label learner (RDFL_SOFT)
classifier trained on retained bags, weights $p_k \cdot c_t$
class centers $\mu_k$ = weighted mean training target
9. Forecast
$\hat{y} = \Sigma_k \hat{p}_k \mu_k$